Arrival · Aspire

I can complete two conditions with three repeats each.

I can record every distance and identify the pattern.

I can write or communicate a conclusion, one improvement and one model limitation.

Point, say, sign, draw, type, write or use an exact scribe. You may pass. Keep your own first answer.

Recall the surface plan. Change one factor, measure distance and keep the other key parts the same.

1. Explain what to do with an unexpected but valid result.
2. Calculate a range and explain how it shows variation.
3. Name one method repair if a run leaves the lane.
4. State one limit of applying the result to a planetary rover.

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Version _____

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Results desk · keep and compare

I can complete two conditions with three repeats each.

I can record every distance and identify the pattern.

I can write or communicate a conclusion, one improvement and one model limitation.

Surface	Run 1 (cm)	Run 2 (cm)	Run 3 (cm)
Smooth	48	51	49
Fabric	27	29	28

Fixed teaching data, not your own new test. Keep 51 cm with the other valid smooth results.

Smooth mean: (48 + 51 + 49) ÷ 3 = 49.33… cm. About 49 cm. Fabric mean: (27 + 29 + 28) ÷ 3 = 28 cm.

Smooth range: 51 − 48 = 3 cm. Fabric range: 29 − 27 = 2 cm. Each smooth value is greater than each fabric value.

Use a release gate to keep the start the same. Test it with the same rover. These runs show one classroom test. They cannot show all real rover conditions.

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Independent · Aspire

I can complete two conditions with three repeats each.

I can record every distance and identify the pattern.

I can write or communicate a conclusion, one improvement and one model limitation.

Point, say, sign, draw, type, write or use an exact scribe. You may pass. Keep your own first answer.

Carry out or direct the fair rover test, record two conditions with three repeats, then give a conclusion, improvement and limitation.

Find each mean and range. Say how the runs vary. How far can this model tell us about a real rover?

Give one change to improve the next test. Say why it helps. Keep the raw results as they are.

Teacher checks the ramp and clear lane. Release without pushing; keep hands, feet and faces away from the moving rover.

Use a safe real test, direct an adult, or use the fixed table. Mark which route you used. Keep the same science decisions.

Choose one route within the full 15 minutes. Keep each raw result.

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My test · prediction and all six results

I can complete two conditions with three repeats each.

I can record every distance and identify the pattern.

I can write or communicate a conclusion, one improvement and one model limitation.

My route: real test / direct an adult / fixed teaching data.

I predict _____ because _____.

Change _____. Measure _____. Keep _____ the same.

Surface	Run	Distance (cm)
Smooth	1	_____
Smooth	2	_____
Smooth	3	_____
Fabric	1	_____
Fabric	2	_____
Fabric	3	_____

If the method fails, keep the note. Repair it, then repeat the run. Note any extra run: _____.

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My test · pattern, conclusion and next method

I can complete two conditions with three repeats each.

I can record every distance and identify the pattern.

I can write or communicate a conclusion, one improvement and one model limitation.

Pattern in all six results: _____

My conclusion and science reason: _____

One method improvement and why: _____

One model limit and why: _____

What the adult heard: _____

My next response, or why I kept it: _____

Keep the first record. A mean or range goes beside the raw results.

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Exit · Aspire

I can complete two conditions with three repeats each.

I can record every distance and identify the pattern.

I can write or communicate a conclusion, one improvement and one model limitation.

Give one method improvement and one model limitation, explaining why each matters.

Use your own result pattern, or name the fixed teaching data. The adult hears your response and names back the science.

What I said: _____

What changed or stayed, and why: _____

Next I will: _____

Point, say, sign, draw, type, write or use an exact scribe. You may pass. Keep your own first answer.

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